Understanding H2S Safety

CNPC Chuanqing Drilling Engineering Company Limited (CCDC)

Defining Lockout & Tagout

LOCK OUT

The process of cutting all sources of energy and installing a personal lock at the source.

The physical lock prevents the starting of machinery while it is being cleaned, maintained, adjusted, or repaired.

TAG OUT

The placement of a tag on an energy-isolating device. It serves as a warning of the danger of operating the machine.

CRITICAL WARNING: A tag indicates 'DO NOT START'. However, a tag Never replaces a lock

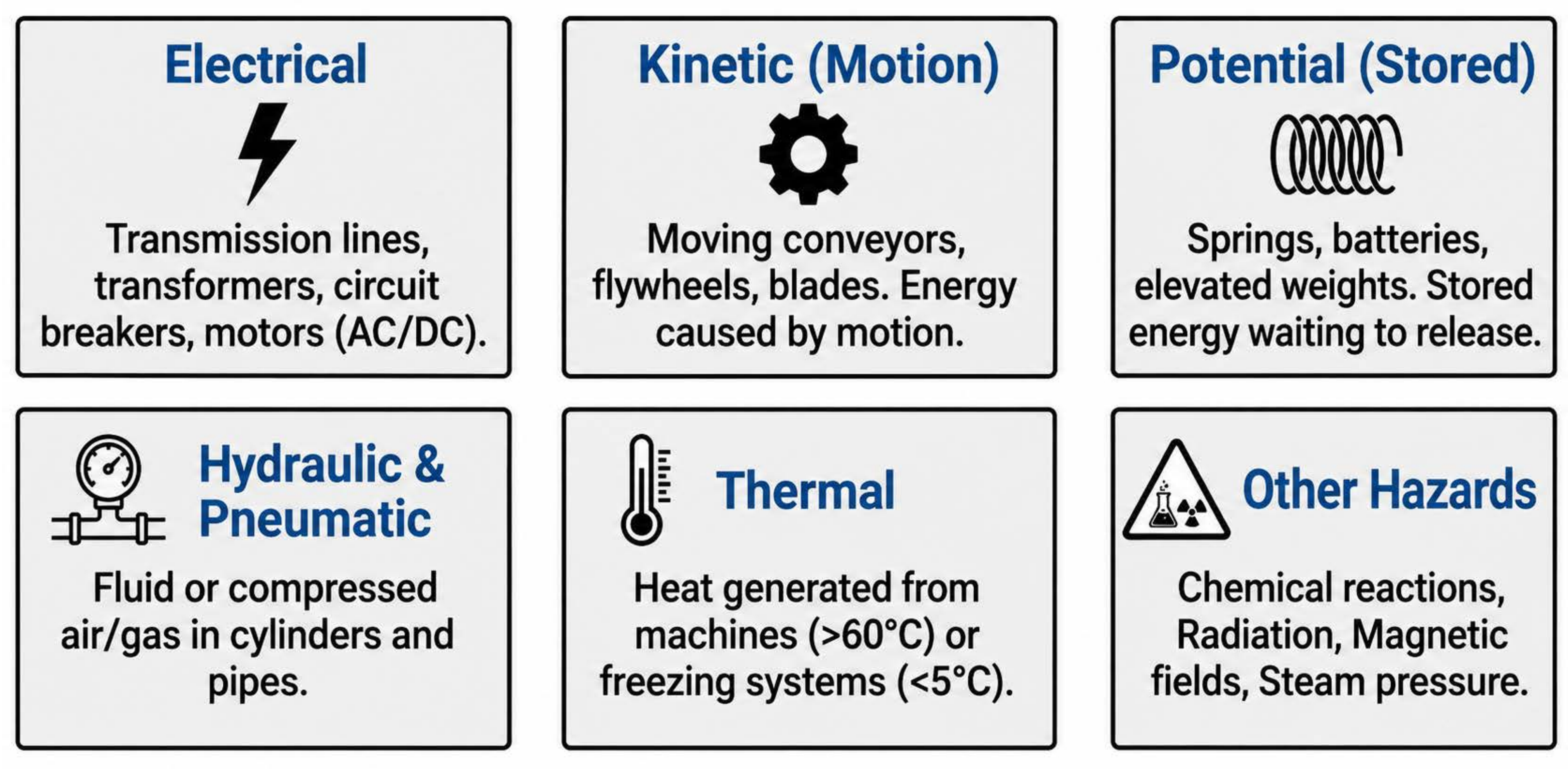
The Objective : Zero Energy State

Definition: The point at which equipment has been safely isolated from ANY chances of re-energization.

Scope: This includes the release of residual(stored) energy, not just disconnecting power.

LOTO is the tool ; **Zero Energy** is the goal

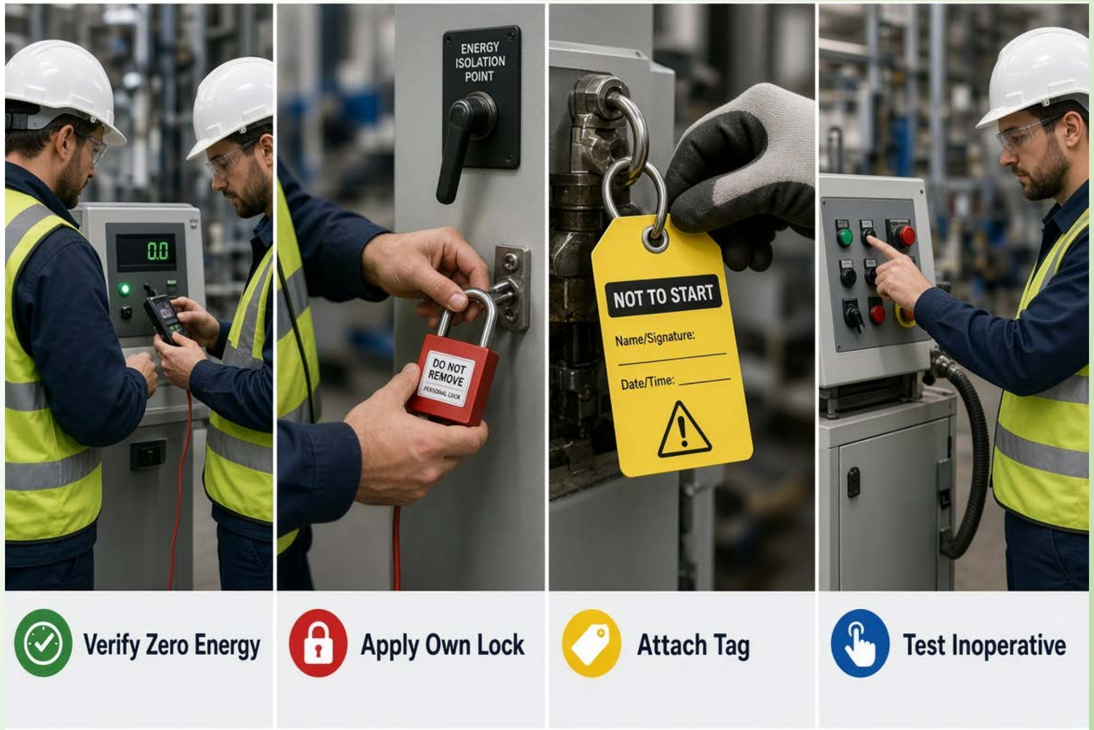
Know Your Enemy: Hazardous Energy Types



Roles & Responsibilities

Employee (LOTO user) Responsibilities

- Verify a competent person to achieved Zero Energy State.
- Lock out the machine using the assigned safety lock(never borrow).
- Apply a tag with: 'Not To Start’,Name/Signature, and Date/Time..
- Test that the machine is inoperative before working.



The 6 Steps of LOTO

Step-1: Preparation & Planning

Understanding the problem

- What needs fixing?
- What hazards are involved?
- Is the equipment-specific LOTO procedure available?

Plan the Work

- Are tools & personnel available?
- Have affected employees been notified

Prepare the Area

- Is the area clear
- Have all disconnect points been located

Step-2 & 3: Stop & Disconnect

Step 2: SHUTDOWN

Use established normal stopping Procedures (e.g. Stop Button, Power Switch)

Involve the worker who normally operates the equipment

Rule: If at a customer site,the customer must shut down the equipment

Step 3: ISOLATE

Disconnect from energy sources (close valves, open breakers).

Wear appropriate PPE

ARC FLASH SAFETY

When opening an electrical disconnect, stand to the RIGHT HAND SIDE of the panel. Use your LEFT HAND to pull the breaker. This protects your body if an arc occurs.

Step-4: Application of Locks & Tags

Rules:

- Use only approved locks & Tags
- Never lend or borrow keys
- Tags must include :

  I. Warning (“Do not start”)

  II. Name

  III. Signature

Step-5: Control Stored Energy

Isolation is not enough. You must neutralize residual energy

Mechanical Motion : Stop motion completely

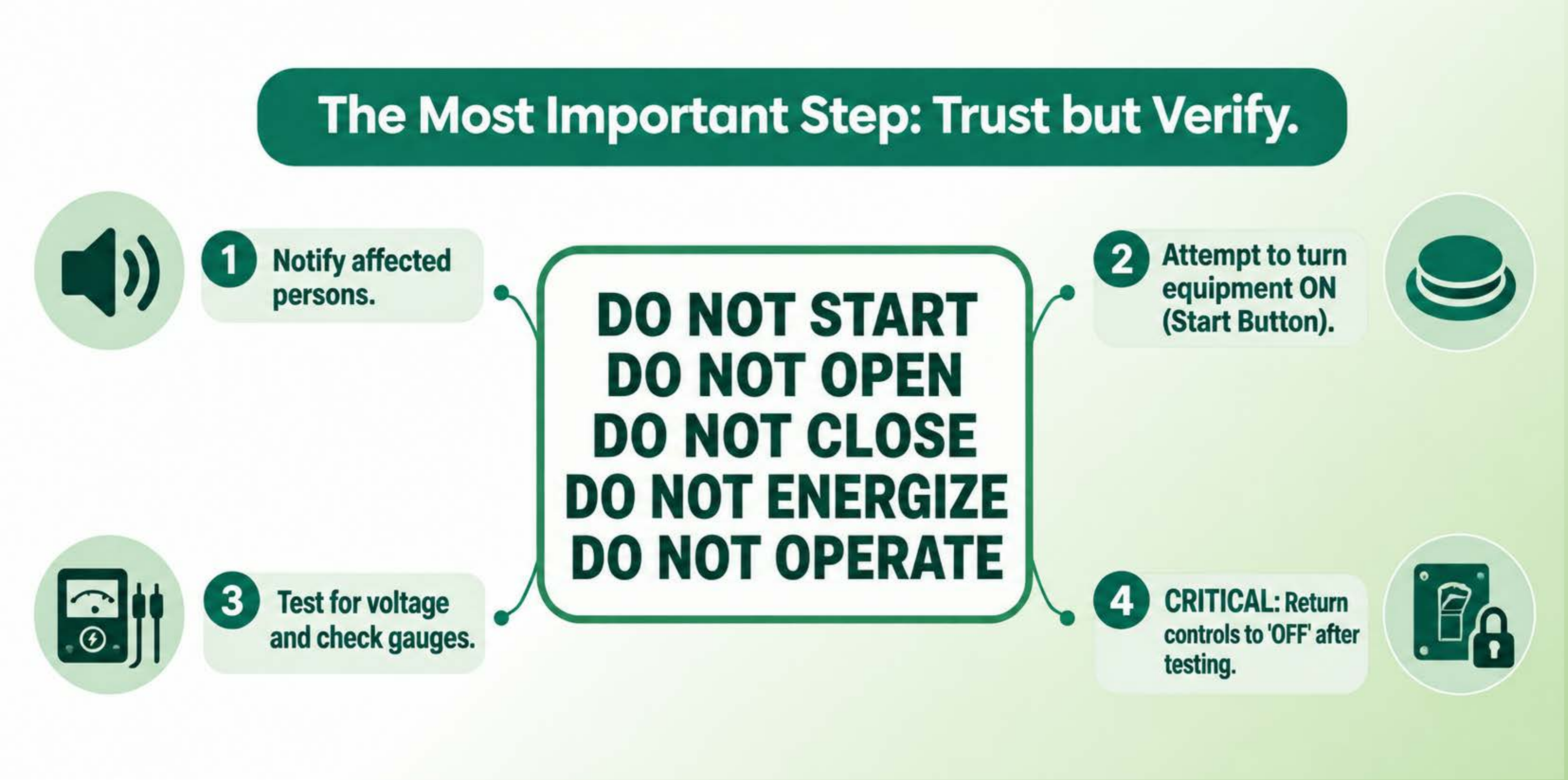
Gravity : Use blocks & pins for elevated parts

Thermal: Allow equipment to cool down or heat up to safe levels

SPRINGS: Block, tag,& lock compressed springs.

Electrical: Drain power from capacitors.

Step-5: Verify Zero Energy



Re-Energizing: The Release Procedure

INSPECT: Verify the is safe. Remove tools. Ensure guards are back in place

NOTIFY: Tell all employees the system is going live

UNLOCK: Remove your own lock & tag

RESTORE: Return system functions.

TEST: Re-energize & test equipment

Avoid These Fatal Excuses

It will only take a minute….

Its lunch time, no one is around

The machine is stopped, it does not need a lock

Always Remember

ONE KEY, ONE LOCK, ONE LIFE

Never lend your key, never remove another worker's lock, and never assume a system is safe without testing it yourself.

Zero Energy, Zero Accident